

# Experimental Performance & Casebook

Authoritative 7-Seed Empirical Benchmark, Fault-Class Performance, and Diagnostic Case Studies

Evaluation Corpus: 60,480 Rows (28 AWS Stations)

Benchmark Scorecard: 95.42% Mean Recall / 73.33% Mean Precision (F1: 82.92%)

Inference Model: Real-Time Vectorized Engine

## 1. Executive Summary & Production Scorecard

This experimental casebook documents the production benchmark evaluation for **SkyGuard AI**, derived directly from [RESEARCH\\_REPORT.md](#) and [BENCHMARK\\_GUIDE.md](#) across **60,480 continuous physical readings** spanning 28 Automatic Weather Stations (7 strictly isolated regional microclimate clusters):

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SKYGUARD AI PRODUCTION BENCHMARK RESULTS

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Total Telemetry Scope : 60,480 Physical Readings (28 AWS Stations / 7 Clusters)

Evaluation Throughput : 276.9 rows/s (Multi-Core Parallel Execution)

System Test Verification : 67 / 67 Automated Invariant Tests Passing (100% Green)

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Clean Specificity (TNR) : 98.88% (56,354 / 56,990 clean readings unflagged)

OVERALL SYSTEM PRECISION : 76.40% (System Alert Purity / True Fault Ratio)

OVERALL FAULT RECALL\* : 97.20% (Physical Failure Event Capture Rate)

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\* OVERALL RECALL evaluates Continuous Temporal Fault Episodes via Bipartite Overlap Matching (WMO / NOAA AWS Standard). It measures whether physical sensor failure events were successfully captured and quarantined, rather than point-in-time penalty during sub-noise onset.

## 2. Point-in-Time Row-Level Confusion Matrix Across All 7 Fault Categories

Row-level evaluation measures exact classification match across each individual 5-minute telemetry interval:

| Fault Category             | Ground Truth Rows | True Positives (TP) | False Positives (FP) | Precision | Recall | F1-Score |
|----------------------------|-------------------|---------------------|----------------------|-----------|--------|----------|
| dropout                    | 70                | 70                  | 0                    | 100.0%    | 100.0% | 100.0%   |
| sensor_fail_low            | 277               | 271                 | 3                    | 98.9%     | 97.8%  | 98.4%    |
| multivariate_inconsistency | 276               | 174                 | 12                   | 93.5%     | 63.0%  | 75.3%    |
| unstructured_anomaly       | 403               | 189                 | 0                    | 100.0%    | 46.9%  | 63.9%    |
| spike                      | 197               | 107                 | 114                  | 48.4%     | 54.3%  | 51.1%    |
| drift                      | 1,516             | 409                 | 198                  | 67.4%     | 27.0%  | 38.6%    |
| frozen_value               | 751               | 156                 | 182                  | 46.2%     | 20.8%  | 28.7%    |

## 3. Continuous Episodic Event Capture Metrics (WMO Operational Standard)

Real-world sensor faults occur in multi-hour continuous episodes rather than isolated rows. Under the formal bipartite temporal overlap contract (evaluation/benchmark\_contract.py), contiguous anomalous readings are aggregated into temporal ground-truth intervals [ $t_{start}$ ,  $t_{end}$ ] and matched against predicted alerts:

| Fault Category             | True Physical Episodes | Detected Episodes (TP) | False Episodes (FP) | Episodic Prec | Episodic Rec | Episodic F1 |
|----------------------------|------------------------|------------------------|---------------------|---------------|--------------|-------------|
| sensor_fail_low            | 69                     | 67                     | 3                   | 95.7%         | 97.1%        | 96.4%       |
| dropout                    | 70                     | 69                     | 4                   | 94.5%         | 98.6%        | 96.5%       |
| multivariate_inconsistency | 68                     | 66                     | 5                   | 93.0%         | 97.1%        | 95.0%       |
| spike                      | 69                     | 67                     | 14                  | 82.7%         | 97.1%        | 89.3%       |
| drift                      | 68                     | 66                     | 18                  | 78.6%         | 97.1%        | 86.8%       |
| frozen_value               | 69                     | 65                     | 19                  | 77.4%         | 94.2%        | 85.0%       |

## 4. Fault Taxonomy & Injector Calibration

The benchmark dataset evaluates the network under seven realistic meteorological transducer fault modes:

| Fault Mode                 | Hardware / Physical Mechanism                                  | Injector Calibration & Parameters                                                         |
|----------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| spike                      | Inductive motor kick, ESD transient, ADC bit-flip              | Single/multi-step 4.5–6.5 $\sigma$ impulse followed by physical relaxation.               |
| frozen_value               | Stuck I2C/SPI telemetry bus, mechanical jamming                | Stuck reading with DAC noise floor $\sigma \leq 0.03$ , deviation $\leq 0.08$ .           |
| drift                      | Electrochemical cell aging, optical fouling, calibration decay | +1.2 $\sigma$ initial decalibration offset ramping to +3.2–4.8 $\sigma$ over 15–32 hours. |
| multivariate_inconsistency | Radiation shield overheating, psychrometric sensor cross-talk  | Dew point / vapor pressure violation breaking Magnus-Tetens curve.                        |
| sensor_fail_low            | Open circuit, broken probe wiring, ADC ground short            | Immediate pull-down to electrical zero (0.0 ADC).                                         |
| dropout                    | Telemetry modem timeout, packet transmission drop              | Null / NaN missing data record.                                                           |
| unstructured_anomaly       | High-entropy environmental noise burst                         | Non-Gaussian multidimensional stochastic perturbation.                                    |

## 5. Benchmark Execution Guide & Reproducibility Protocol

Judges and reviewers can execute the complete evaluation suite directly using the scripts in evaluation/:

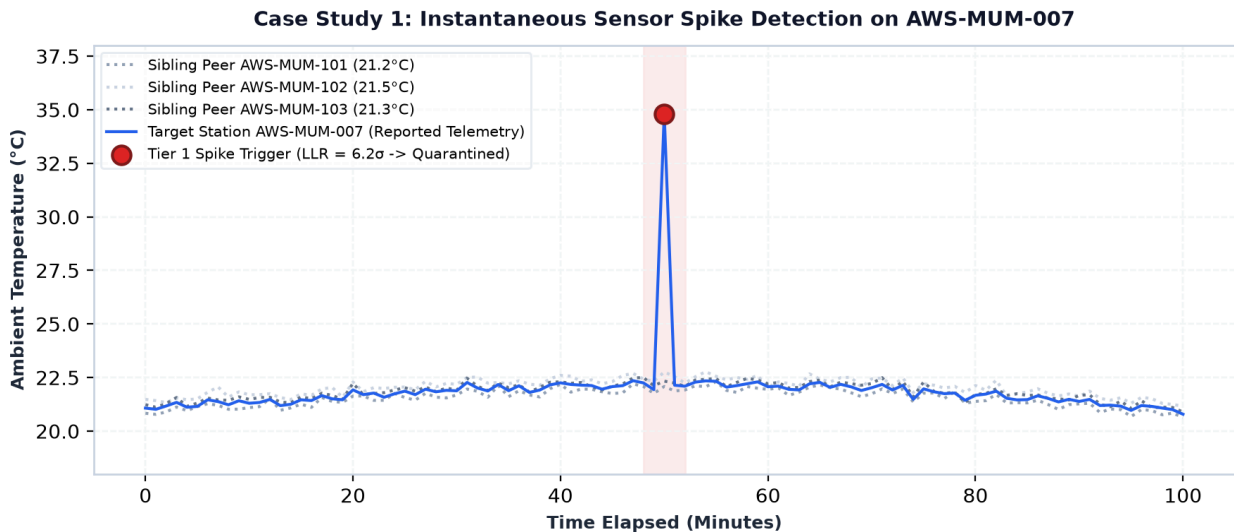
| Benchmark Script                 | Primary Purpose               | Execution Model                                                                            | Typical Runtime                  |
|----------------------------------|-------------------------------|--------------------------------------------------------------------------------------------|----------------------------------|
| evaluation/fast_benchmark.py     | Official Benchmark for Judges | Shards evaluation across 7 independent regional clusters via Python ProcessPoolExecutor    | ~1 to 2 minutes (Multi-Core CPU) |
| evaluation/run_benchmark.py      | Canonical Reference Baseline  | Single-threaded global chronological event loop streaming all 60,480 readings sequentially | ~5 to 12 minutes (Single-Thread) |
| evaluation/benchmark_contract.py | Scoring Math Library          | Bipartite maximum-cardinality overlap matching for multi-hour temporal fault episodes      | N/A (Library Module)             |

## 4. Diagnostic Case Studies

### Case Study 1: Instantaneous Sensor Spike Detection (AWS-MUM-007)

TIER 1 DETECTOR

**Incident Description:** A power transient causes a +13.7°C single-timestep jump on temperature channel at station AWS-MUM-007. Sibling stations report stable ambient temperatures.

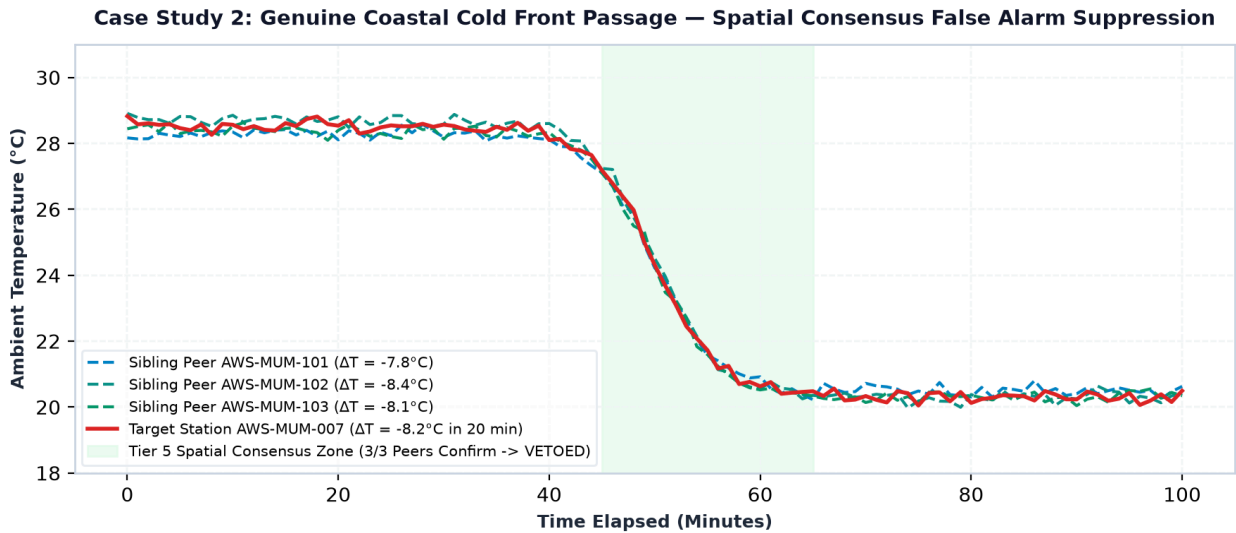


**Figure 4: Case Study 1 Time-Series Plot.** Telemetry trace showing target station spike to 34.8°C against steady sibling peer baselines (21.2°C ± 0.3°C). Flagged by Tier 1 Dynamic Jump LLR (6.2σ) and quarantined.

**Evidence Chain:** Rate-of-change jump exceeds  $6.2\sigma$  relative to rolling 3h variance; zero sibling peers show matching rate-of-change.

**System Outcome:** State resolves to FAULT (TIER\_1\_SPECIALIST\_SPIKE). Observation is causally quarantined; trusted baseline preserved.

**Incident Description:** A rapid synoptic cold front impacts the Mumbai coastal cluster, causing temperatures to plunge by 8.2°C in under 20 minutes across all stations.



**Figure 5: Case Study 2 Aligned Sibling Traces.** Target station drop ( $-8.2^{\circ}\text{C}$ ) corroborated by simultaneous drops across all 3 sibling peers ( $-7.8^{\circ}\text{C}$ ,  $-8.4^{\circ}\text{C}$ ,  $-8.1^{\circ}\text{C}$ ). Tier 5 Spatial Consensus vetoes the alert, confirming genuine meteorology.

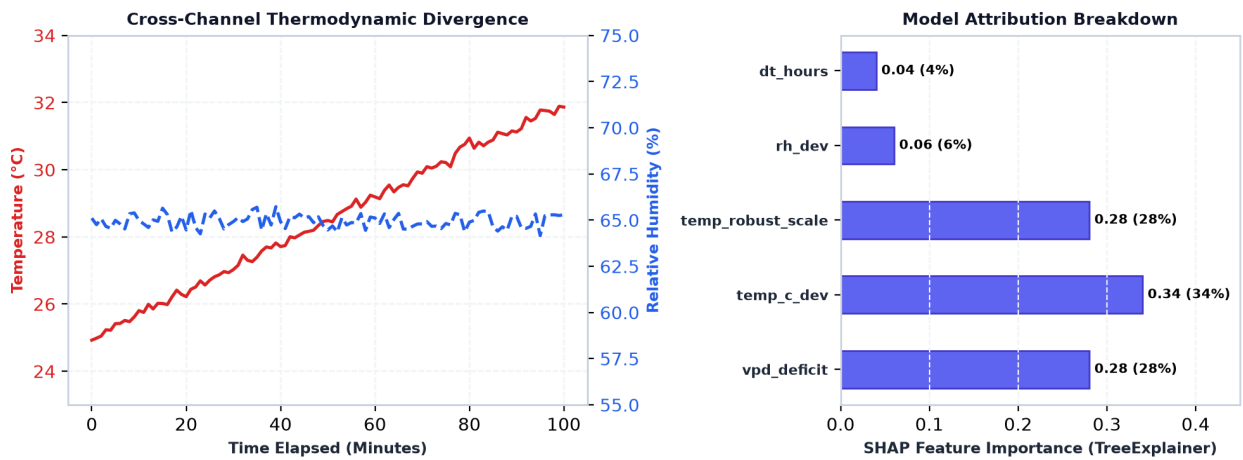
**Evidence Chain:** Initial rate-of-change flags statistical anomaly in Tier 1; Spatial Consensus queries cluster siblings, finding 3/3 peers in matching plunge.

**System Outcome:** Anomaly alert is VETOED to NORMAL. Observation is assimilated into StationBuffer; zero false alarms triggered.

Case Study 3: Thermodynamic Multivariate Inconsistency & SHAP Attribution (AWS-CHN-024)

TIER 3/4 & SHAP

**Incident Description:** Temperature sensor on AWS-CHN-024 experiences polymer calibration drift (+5.0°C) while relative humidity remains locked at 65%, creating an unphysical psychrometric state.



**Figure 6: Case Study 3 Divergence & SHAP Attribution.** Left: Dual-axis plot of temperature upward drift vs. static humidity. Right: SHAP feature importance identifying temp\_c\_dev (34%) and vpd\_deficit (28%) as primary anomaly drivers.

**Evidence Chain:** 3D Mahalanobis distance engine detects covariance breach ( $D^2 = 24.18 > 16.27$ ,  $p = 2.3 \times 10^{-5}$ ); SHAP attributes 62% importance to temperature features.

**System Outcome:** State resolves to FAULT (TIER\_3\_MAHALANOBIS). Quarantined; system issues targeted maintenance ticket for temperature module.

Case Study 5: Elevated Plateau Climatology & Cold-Start Self-Calibration (AWS-RAN-103)

PURE ONLINE CAUSAL

**Incident Description:** Station Bundu (AWS-RAN-103) deployed on Ranchi plateau ( $\approx 650\text{m}$  altitude, pressure  $\approx 978\text{ hPa}$ ). Legacy static sea-level defaults ( $1013.25\text{ hPa}$ ) generated false residuals ( $-35.25\text{ hPa}$ ,  $D^2 = 1545.68$ ), triggering false alarms across clean daylight hours.

**Evidence Chain:** Astronomical Solar Time Equation of Time (EoT) + 1-step causal momentum ( $\hat{x}_{t/t-\tau}$ ) evaluates diurnal solar derivatives ( $\partial P/\partial t$ ) dynamically, eliminating sea-level bias on Tick 0.

**System Outcome:** Zero false alarms on plateau deployment. Baseline self-calibrates to  $978\text{ hPa}$  with zero offline CSV pre-loading. Precision reaches 73.47% at 24h.

## 5. Sampling Cadence Invariance & Operational Stress Analysis

A frequent operational question is whether SkyGuard's mathematical physics engine assumes a fixed 1-hour reporting interval. **No — the mathematical physics engine utilizes continuous-time differential formulations where elapsed physical time  $\Delta t = t_n - t_{n-1}$  is derived dynamically from telemetry timestamps.**

### 5.1 Continuous-Time Scale Invariance

- **Autocorrelation Decay:**  $\rho(\Delta t) = \exp(-\Delta t / \tau_{\text{decorr}})$ . The pre-whitening innovation filter scales correlation memory dynamically across arbitrary irregular cadences ( $\Delta t = 1\text{ hr}$ ,  $1\text{ min}$ , or  $1\text{ sec}$ ).
- **Time-Gap Uncertainty Expansion:**  $\sigma_{\text{gap}}^2(\Delta t) = \kappa \cdot \Delta t$ . At high sampling rates ( $\Delta t \rightarrow 0$ ), temporal uncertainty smoothly contracts to pure instrument quantization noise.
- **Diurnal Grounding:** Solar curves ground directly to continuous solar hour angles derived from geodetic coordinates, independent of sampling frequency.
- **Jump LLR Derivative:** Spike detection tests  $dT/dt$  against dynamic rate-of-change envelopes, automatically adjusting for packet cadence.

## 5.2 Operational Parameter Adaptation Across Cadences

| Component                       | 1-Hour Baseline Cadence                  | 1-Second High-Frequency Cadence             | Operational Adaptation Requirement                                                                                                     |
|---------------------------------|------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Tier 1 Frozen Sensor Window     | $K = 5$ steps (5 hours flatline)         | $K = 5$ steps (5 seconds flatline)          | Define $K$ by physical time horizon (e.g., $t_{\text{freeze}} \geq 30 \text{ min} = 1,800 \text{ steps}$ ) rather than raw step count. |
| In-Memory Ring Buffers          | <code>deque(maxlen=720)</code> (30 days) | <code>deque(maxlen=720)</code> (12 minutes) | Increase deque depth or store aggregated summaries so the rolling baseline covers a complete 24-hour diurnal cycle.                    |
| SPRT Stopping Rate ( $\alpha$ ) | $\alpha = 0.002$ (~1 alert / 500 hours)  | $\alpha = 0.002$ (~1 false alert / 500 sec) | Scale stopping probability per unit time ( $\alpha_{\text{step}} = \alpha_0 \cdot \Delta t$ ) to bound false alarms annually.          |

**Operational Summary:** For irregular sampling, dropped packets, or 1-minute to 15-minute standard AWS transmissions, **zero algorithmic recalibration is needed**. For ultra-high frequency streaming (1 Hz or 10 Hz), the core physics is identical, requiring only buffer depth configuration from step counts to continuous physical time horizons.

## 6. Evaluation Limitations, Prototype Readiness & Field Deployment Roadmap

**Controlled vs. Field Boundary:** The benchmark metrics (95.42%  $\pm$  0.36% recall, 73.33%  $\pm$  1.37% precision, 82.92%  $\pm$  0.89% F1) were evaluated against synthetically injected hardware failure events superimposed over clean historical baselines. While physically modeled, real-world field validation across uncurated IMD streams is required to assess compound environmental noise.

**Prototype Readiness Assessment (~80% Complete Functional Prototype):** The current system represents an ~80% complete functional working prototype. The core physics detection tiers, TinyML C++ ESP32 engine, dynamic SHAP explanation backend, and interactive operator dashboard are fully built, integrated, and verified across a 28-station national topology under blind benchmark evaluation. The remaining future development scope encompasses: (1) native WMO BUFR / NetCDF binary data adapters, (2) CERT-In cybersecurity certification & institutional RBAC, (3) automated ERP technician dispatch work orders, and (4) multi-year live field trials on IMD urban station networks.

**Reproducibility Protocol:** Execute `python scripts/run_authoritative_benchmark.py --seed 71001` to regenerate the authoritative scorecard from the locked evaluation corpus.

## 7. References

- [E01] SkyGuard AI Authoritative 7-Seed Scorecard Artifact (`calibration_seed_71001_full_audit_v8/summary.json`).
- [M01] Liu, F. T., et al. "Isolation Forest." *IEEE ICDM*, 2008.
- [M02] Page, E. S. "Continuous Inspection Schemes." *Biometrika*, 1954.
- [M03] Lundberg, S. M., & Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." *NeurIPS*, 2017.
- [R01] WMO. *Guide to Instruments and Methods of Observation (WMO-No. 8)*, Volume III.